Homework 3: Symbolic Task Planning and Grasping

16-662: Robot Autonomy

Spring 2026

Due: Wednesday, March 18, 2026 at 11:59 PM ET

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Introduction

This homework will focus on **task planning** and **grasping**. We will be defining and solving problems pertaining to the arrangement of blocks using symbolic planning framework we learnt in class. We will also analyze grasp feasibility from a mathematical modeling point of view. This could be a challenging assignment, please start early!

AI Usage Policy

The use of AI to generate code for this homework is **strictly forbidden**. You are instead recommended to attend the Teaching Assistants' office hours or research by yourself or with your peers.

Part 1: Symbolic Planning

This problem set introduces two files: **SymScript.py** and **SymPlanner.py**. SymPlanner includes code for a domain-independent symbolic planner. SymScript defines specific domains and queries. We will be updating both files using concepts we learnt in class.

1.1 Block World Domain

Start by examining the make_Room_domain() function in SymScript.py. It specifies actions using the ActionSchema() class in SymPlanner.py that can be used with the Planner() class functions.

Similarly, define actions in the **make_Block_domain()** function that can be used to solve problems in the Block World. Think about the actions that need to be performed to move blocks around like you did in Lab1.

NOTE: You can't solve the problem yet because we haven't defined a way to plan about the states and actions!

1.2 Discrete Search

Next, you will implement A* search to find a plan of actions to solve the problem using the Planner() class functions. The problem has been setup in the **plan()** function. You must fill in the remaining code.

Once you have successfully implemented the search algorithm, you will be able to solve the Room problem by running SymScript.py.

1.3 Planning In Block World

Now that you have defined the action space for the Block World and you have written a working discrete search planner about states and actions, you can now solve problems in the Block World!

Write a block of code in the main function of SymScript.py using the Room problem block as an example to solve the Block problem. You will use the action domain you have defined in part 1.1 and the planner in part 1.2.

1.4 Another Block World Problem

We are now going to create a new problem for the Block World domain. The initial and goal states are shown in Fig. 1.

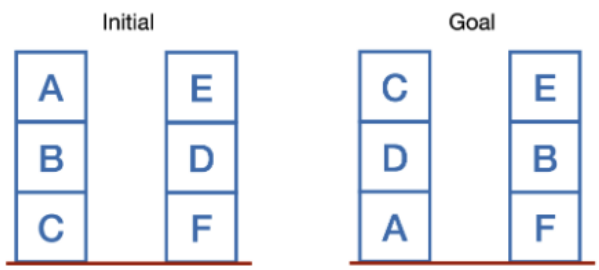

Figure 1: Another Block World Problem

Add the initial and goal states to **make_Block2_problem()** in SymScript.py. Solve the problem in the same way you solved the first two problems.

1.5 Submission

Submit your **SymPlanner.py** and **SymScript.py** files along with **screenshots of the plans** generated for the two Block World problems in your writeup.

Part 2: Grasping

2.1 Removing Contacts

Fig. 2 shows a 2D (X, Y, θ) example of a form-closure grasp with six frictionless contacts. Write out the grasp matrix $\mathbf{G}$ for this grasp. Plot the grasp vectors $\mathbf{g}$ using an F_X vs. T_θ plot and an F_Y vs. T_θ plot.

In our lecture example, we found a solution using only four contacts to achieve a form-closure grasp of a box. Without shifting any of the contact points, can any of the contact points be removed while still maintaining form closure? Explain your answer in 1–2 sentences.

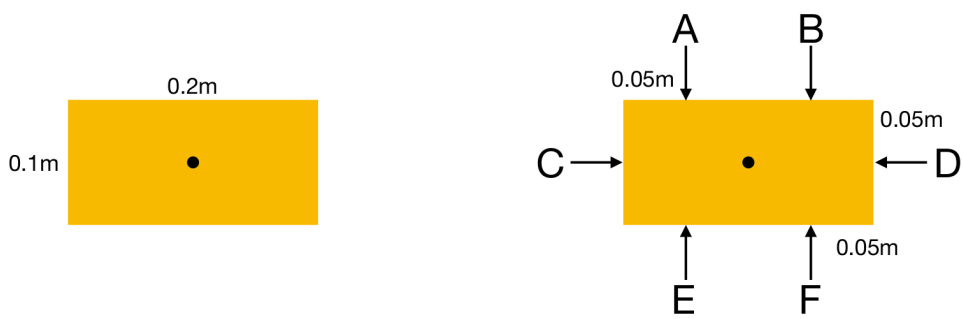

Figure 2: Object and grasp for problem 2.1

2.2 Force Closure Synthesis

Consider the triangle shape in Fig. 3. You need to choose one of the three possible grasps (A, B, or C) to execute. Each grasp consists of two contacts on the slanted edges (e.g., B1 and B2 for the B grasp) with a coefficient of Coulomb friction of 1, as well as a shared contact at S that is frictionless. Grasp A contacts are at the top corner, grasp C contacts are at the bottom corners, and grasp B contacts are exactly in the middle of the sides. For each grasp, write out the grasp matrix $\mathbf{G}$ and compute the grasp quality metric $\sqrt{\det(\mathbf{G}\mathbf{G}^T)}$. Based on this grasp metric, which contacts provide the best grasp A, B, or C? Explain your answer in 1-2 sentences.

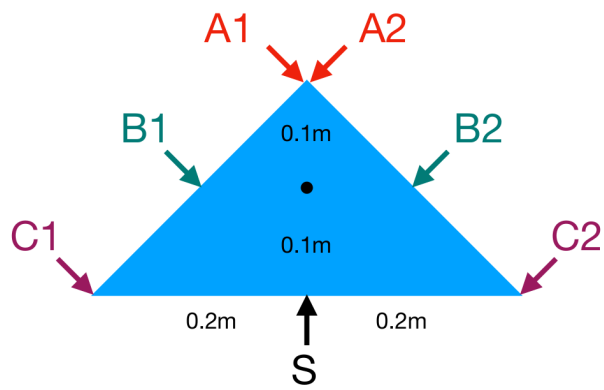

Figure 3: Objects and grasp for problem 2.2

Submission Instructions:

- The submitted zip file should unzip into a zip file named `andrewid_hw3_code_16662.zip` AND one pdf named `andrewid_hw3_16662.pdf` with your solutions.
- Please ensure your venv or conda environment is NOT submitted in the zip folder.
- Kindly upload your zip file respecting the following **naming convention**: `andrewid_hw3_16662.zip`

That's it! You have completed HW 3! =D